Unit 2: Solution Techniques (The "Toolbox")

In this unit, we explore the specific mental and analytical strategies used to break down complex problems. These techniques allow a problem solver to move from a vague challenge to a structured solution.

2.1 Cognitive and Creative Strategies

These methods involve changing how we perceive a problem to make it easier to handle.

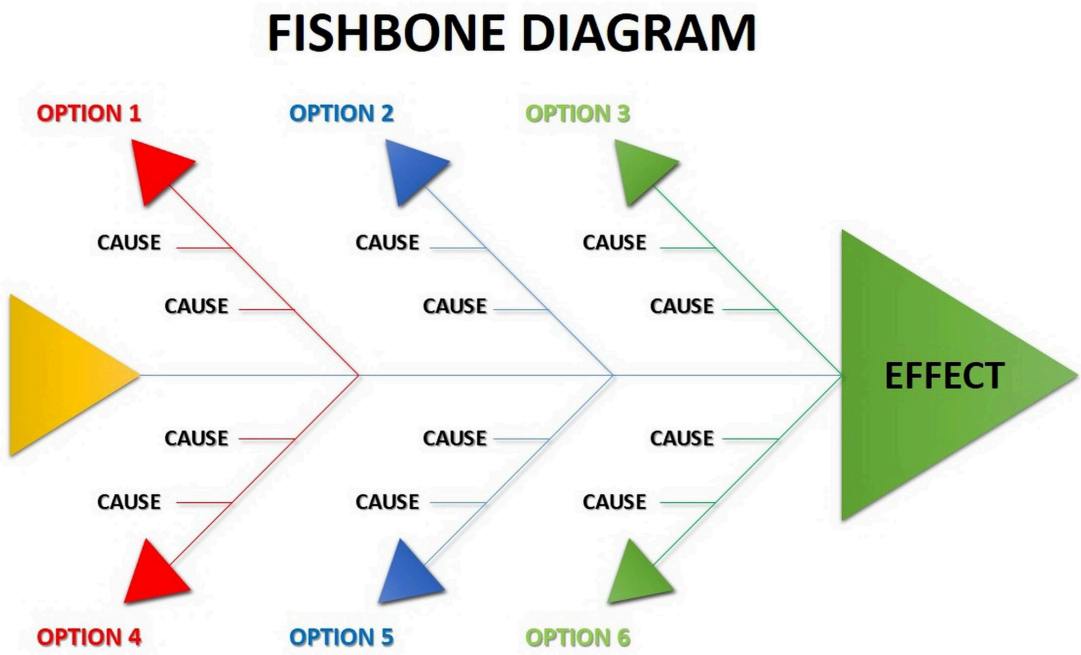
- **Abstraction:** The process of removing unnecessary details to focus on the core features of a problem. In computing, this means focusing on *what* a system does rather than *how* it is implemented.
- **Analogy:** Finding a similar problem that has already been solved and applying its solution to the current situation.
- **Brainstorming:** A group or individual technique for generating a large number of ideas or potential solutions without immediate criticism.
- **Lateral Thinking:** Approaching a problem from an indirect or creative angle rather than following a traditional step-by-step logic. It is often described as "thinking outside the box."

2.2 Analytical and Systematic Strategies

These are more structured methods used to narrow down the cause of a problem or find a technical fix.

- **Trial and Error:** A fundamental method where various attempts are made until a successful one is found. While simple, it can be inefficient for complex computing problems.
- **Hypothesis Testing:** Formulating an explanation for a problem (a hypothesis) and then conducting experiments to prove or disprove it.
- **Root Cause Analysis (RCA):** A method of problem-solving used for identifying the "root" of faults or problems. A common tool here is the **"5 Whys"**

technique—repeatedly asking "Why?" until the underlying cause is revealed.



- **Means-End Analysis:** Identifying the difference between the current state and the goal state, and then taking actions (means) to reduce that difference step-by-step.

The **Fishbone Diagram** (also known as the **Ishikawa Diagram**) is a primary tool used in **Root Cause Analysis (RCA)**, which we covered in section 2.2.

In the context of problem solving, it provides a visual way to brainstorm and categorize the potential causes of a specific problem (the "effect"). For computing students, it is particularly useful for debugging complex systems where a failure might stem from multiple sources like hardware, software logic, data input, or user error.

Usefulness of the Fishbone diagram:

1. **Structural Brainstorming:** Unlike a list, the fishbone structure forces you to think about different categories of causes.
2. **Visualizing Relationships:** it shows how various factors might interact to create a "bottleneck" or a system crash.
3. **Path to Solution:** Once the "bones" (causes) are identified, the student can apply **Hypothesis Testing** to see which cause is the true culprit.

2.3 Advanced Structural Strategies

These techniques are heavily used in computer science and software engineering.

- **Divide and Conquer:** Breaking a large, complex problem into smaller, more manageable sub-problems, solving them individually, and then combining their results.
- **Reduction:** Transforming a difficult, unknown problem into a problem we already know how to solve.
- **Morphological Analysis:** Breaking a system down into its basic functions and exploring all possible ways to achieve each function to find a novel combination.
- **Method of Focal Object:** Synthesizing the characteristics of different, seemingly unrelated objects to create something new or solve a design challenge.

2.4 Logical and Proof-Based Strategies

- **Proof:** Using logical deduction or mathematical evidence to demonstrate that a solution is correct and will work under all specified conditions.
- **Research:** Consulting existing literature, documentation, or technical forums to identify how others have handled similar challenges.